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Strategic Asset Allocation · Asset Management Challenge 2026 · Round 1

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Investment policy (Checkpoint 1)

BlackHawk Capital allocates USD 100 million across eight equity, bond and commodity ETFs, combining geographic diversification with a US value tilt and setting the weights through Capped Equal Risk Contribution. The aim is to participate in growth while spreading risk across equity markets, Treasury duration, short corporate credit, gold and commodity futures.[14]

Our objective is the Sharpe Ratio on which the challenge is evaluated, under the 8% volatility target the guidelines set, within the AMC range of 7-9%. Using the latest 60 complete monthly USD returns, October 2021-September 2026, the allocation has estimated volatility of 8.58% and a one-month 5% normal VaR of 4.07%, or USD 4.07 million; VaR describes a loss quantile and losses can exceed it. We sit in the upper half of the band because the capped solution lands there on its own; cash earns no interest here, so de-risking to 8.00% would cost return without improving the ratio.[14]

Table 1. Strategic capital allocation and normalized risk contribution

ETF	Capital weight	Capital USD m	Risk contribution
VOO	9.91%	9.91	14.04%
VTV	10.54%	10.54	14.16%
IEFA	8.47%	8.47	13.95%
EEM	8.30%	8.30	13.93%
IEF	24.04%	24.04	14.79%
IGSB	12.00%	12.00	3.50%
GLD	11.75%	11.75	14.15%
DBC	15.00%	15.00	11.48%

Source: BlackHawk Capital, own calculations and compilation. [14]

We hold long positions without leverage, short selling or inverse/leveraged ETFs. Our own limits are 50% equity, 12% IGSB, and 15% each in GLD and DBC. The IGSB cap limits reliance on one short-credit instrument and on its estimated covariances; the real-asset caps limit exposure to each distinct economic risk. The equity ceiling preserves room for the other sleeves. These limits express our concentration policy rather than correcting the covariance estimates themselves; Section 4.3 sets out the estimation-error reason that makes such a policy necessary.[14]

Maximum acceptable drawdown: 20% below the portfolio's running wealth peak. This is our stated loss tolerance, set above the observed 13.10% fixed-allocation and 14.19% net walk-forward drawdowns. It is not a guaranteed loss limit. The selected strategy has no dynamic drawdown-control overlay; AMC positions remain unchanged between rounds. USD cash earns no interest. Final share quantities use Bloomberg closing prices of 16 October 2026.[14][13][23]

No single position carries more than 14.8% of total risk, so none approaches half the risk budget; the four equity funds together account for 56.1%, and that concentration is intended. Capital weights are rounded for presentation; the exact weights sum to 100%.[14]

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CONTENTS

1 Market outlook 3

2 Portfolio construction 6

3 Data and portfolio inputs 7

4 Allocation methodology 8

5 Portfolio diversification 13

6 Backtest and historical risk 16

7 Alternative allocations and robustness 21

8 Costs liquidity and implementation 25

Appendix A Instrument identifiers and sources 28

Appendix B Code and data map 29

Appendix C Research configurations 30

References 31

1 Market outlook

1.1 Growth inflation and the policy setting

Our starting view is that growth remains investable, while inflation and interest rates deserve a meaningful place in portfolio construction. The IMF projects global growth of 3.0% in 2026 and 3.4% in 2027. Energy pressures and strong technology demand make the regional picture uneven. We want participation in corporate earnings without making the whole fund depend on an uninterrupted equity rally or an immediate easing cycle.[5]

Table 2. Selected official forecasts available at 12 October 2026

Measure	2026	2027	Convention and institution
Global real GDP	3.0%	3.4%	Calendar-year growth; IMF July WEO
US real GDP	2.3%	2.4%	Q4 over Q4; Fed September SEP
Euro area real GDP	0.9%	1.4%	Annual growth; ECB September projections
US headline PCE inflation	3.7%	2.3%	Q4 over Q4; Fed September SEP
Euro area headline HICP	3.0%	2.5%	Annual average; ECB September projections

Source: IMF; Federal Reserve; European Central Bank. [5][2][4]

These projections come from different institutions and use different conventions. Federal Reserve inflation is measured Q4 over Q4; euro-area HICP is an annual average. We use them to frame the conditions our holdings may face. They are not direct forecasts of ETF returns.[2][4][5]

The latest US inflation release available at our 12 October information cutoff is August: headline CPI rose 3.4% year on year, core CPI 2.4%, and energy 16.3%. The September release is scheduled for 14 October. The energy pressure matters to us because it can weaken purchasing power while keeping monetary policy restrictive.[3]

The Federal Reserve raised its target range to 3.75-4.00% in September, and the ECB took its deposit facility to 2.50%. The September Federal Reserve projections also moved above the June path. We therefore retain Treasury duration for its strategic role while recognizing the risk of further yield increases. Our weights are anchored in the risk budget, with the macro view guiding the choice of exposures.[1][2][4]

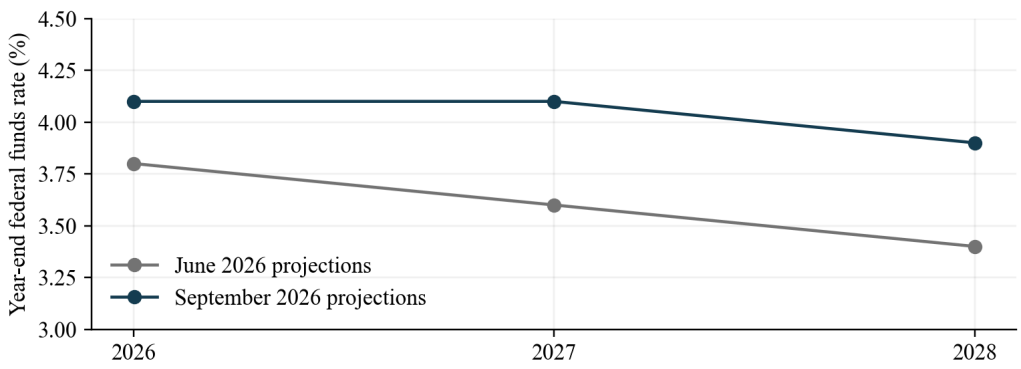


Figure 1. Federal Reserve policy expectations were revised upward

Source: BlackHawk Capital chart of Federal Reserve data. [2][14] BlackHawk Capital chart of Federal Reserve projections: September 2026 versus June medians. Year-end policy-rate forecasts.

After our 12 October information cutoff, the September releases arrived and both exceeded expectations. Headline CPI rose 3.7% over the year, up from 3.4% in August, with core CPI at 2.6% and energy 20.4% higher over twelve months. Producer prices for final demand rose 1.3% in the month, the largest increase since March 2022, and 6.3% over the year. Energy accounts for most of both surprises, so one month does not establish a trend. We record these figures because they are consistent with the inflation and policy risks described above. They change no estimate, weight or historical test in this report: the covariance window, the allocation and the backtests all remain at the 12 October data cutoff, and our weighting rule does not use an inflation forecast.[25][26]

1.2 Equity geography and the value allocation

Our US equity allocation combines the broad S&P 500 with a value tilt. We want access to US earnings and innovation, while reducing reliance on a single valuation and sector mix. The reported P/E snapshots are 26.67 for VOO, 20.09 for VTV, 17.43 for IEFA and 15.97 for EEM. These differences give context to the allocation; the funds also differ in sectors, countries and snapshot dates.[11]

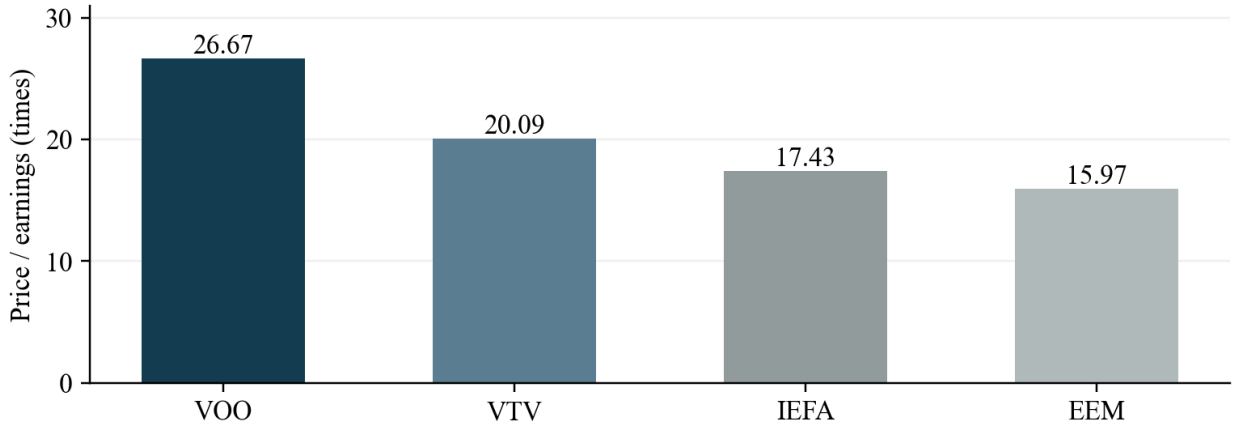


Figure 2. Valuation differences across the equity funds

Source: FactSet. [11] BlackHawk Capital chart of reported P/E snapshots: VOO/VTV 31 August; IEFA/EEM 9 October 2026.

VTV changes the composition of our US exposure, including the balance between technology and financial companies. We accept its overlap with VOO because the purpose is a style tilt within the same market. The current 37.21% equity weight leaves room for other return and risk drivers, while keeping equities important to the fund's long-term outcome.[14]

IEFA broadens the corporate and regional mix across developed markets outside the US, including Japan and Europe. We prefer this broad exposure to a narrow country call based on the latest forecast. USD listing provides a common reporting currency but leaves the underlying currency risks to be assessed.[11][14]

EEM adds emerging-market businesses, with important technology-related exposures. The IMF describes stronger technology demand in Korea; the September country export assigns 28.9% of EEM to Taiwan and 21.4% to South Korea. These concentrations bring opportunities and geopolitical risk. Our 8.30% fund weight limits their effect on total capital, while the correlation analysis shows the equity market risk that remains.[6][12][14]

1.3 Bonds gold and commodity exposure

The bond sleeve separates Treasury duration from short corporate credit. IEF reports effective duration of 6.8 years, versus 2.7 years for IGSB in the August snapshots. IEF can benefit from falling nominal yields and weaker growth; it loses value when yields rise. IGSB reduces rate sensitivity while retaining exposure to corporate credit spreads. We cap IGSB at 12% to prevent one quiet-looking instrument from absorbing too much capital.[11][14]

We combine this nominal-rate exposure with gold and commodity futures because inflation shocks can damage both equities and nominal bonds. The October EIA outlook estimates September Brent at USD 114 a barrel, projects USD 105 for the fourth quarter and USD 84 on average in 2027 as supply conditions normalize. DBC gives us broad commodity exposure, including sensitivity to supply conditions, but the projected normalization is also an adverse case for that sleeve.[7][14]

Gold has a different role. The World Gold Council estimates central-bank purchases of 288.9 tonnes in the second quarter, after a revised 56.5 tonnes in the first. At the same time, gold-backed ETFs experienced roughly 45 tonnes of outflows in Q2. Official demand supports the case for owning bullion; the investor flows remind us that real yields, the dollar and changing demand can still weigh on its price. [8]

The resulting weights are 11.75% in GLD and 15% in DBC. GLD provides physical bullion without futures roll effects. DBC provides a broader commodity basket, with roll effects and a higher fee. We give those costs and the position's execution capacity explicit attention when evaluating whether the exposure is worth holding.[14]

1.4 Recent market performance and allocation implications

Recent returns make the trade-off tangible. Through 9 October, DBC has gained approximately 47.4% in 2026, while GLD has declined about 3.0% and IEF about 4.2%. EEM has outperformed the developed-market equity funds. These are reinvested-dividend ETF returns in USD, not the return of our fund since the challenge began.[14]

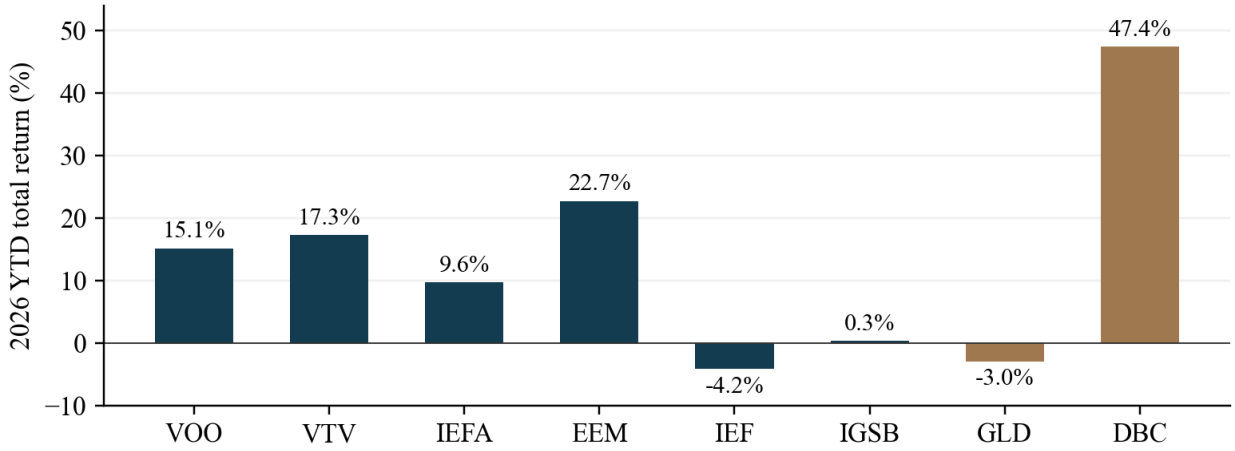


Figure 3. 2026 returns show different behavior within each asset class

Source: BlackHawk Capital, own calculations and compilation. [14] Reinvested-dividend USD returns through 9 October 2026 relative to the last observation of 2025.

We retain commodity exposure for the supply and inflation risks it introduces into the portfolio, while recognizing the cost of entering after a large rally. We retain Treasury duration because its strategic role extends beyond its recent negative return. Neither recent winner nor recent loser determines the weight on its own: the covariance estimate, risk-budget objective and concentration limits work together.[14]

Our view is therefore expressed through the breadth of the allocation, with its size controlled by risk. The portfolio would be challenged by simultaneous equity and bond weakness, a sharp commodity reversal, or a technology correction spreading across developed and emerging markets. The following allocation analysis and stress tests show where we expect those pressures to appear.[14]

2 Portfolio construction

2.1 Asset universe and strategic roles

We want the portfolio to earn its return from several sources. Equities account for 37.21% of capital, bonds for 36.04%, and gold plus broad commodities for 26.75%. The eight positions give us a manageable allocation whose major risks we can explain and monitor.[14]

Our equity choices express geography and investment style. VOO gives us a broad US core; VTV changes its style balance towards value; IEFA adds developed markets outside the US; and EEM brings emerging-market exposure. For Round 1, we prefer these broad building blocks to a series of sector bets. Later tactical decisions will be evaluated against this strategic reference.[14]

Table 3. Eight selected instruments and their strategic roles

ETF	Security	Role
VOO	Vanguard S&P 500 ETF	US equity core
VTV	Vanguard Value ETF	US value style
IEFA	iShares Core MSCI EAFE ETF	Developed equity outside the US
EEM	iShares MSCI Emerging Markets ETF	Emerging equity including South Korea
IEF	iShares 7–10 Year Treasury Bond ETF	Treasury duration
IGSB	iShares 1–5 Year Investment Grade Corporate Bond ETF	Short corporate credit
GLD	SPDR Gold Shares	Physical gold
DBC	Invesco DB Commodity Index Tracking Fund	Broad commodity futures

Source: FactSet. [11]

The bond sleeve separates two risks. IEF holds US Treasuries and supplies nominal duration. IGSB holds short investment-grade corporate bonds and brings credit exposure with less rate sensitivity. GLD adds physical bullion; DBC adds commodity futures and the effects of their forward curves. We assess these exposures together, because an asset-class label alone tells us little about how a position behaves in a stressed portfolio.[14]

The market outlook explains why we want these building blocks. The weighting rule determines how much of each we hold. This distinction keeps a short-term view on growth, inflation or valuations from silently replacing the allocation process.[14]

3 Data and portfolio inputs

3.1 Sources and estimation conventions

The eight price histories come from the team's own price exports; Appendix A documents each field and its source. We use Gross Total Return for performance, incorporating the dividend-reinvestment interpretation confirmed by the team, and raw closing Price for share quantities. The common history provides 167 complete monthly returns from November 2012 to September 2026. October is incomplete and excluded. All eight funds have the final 60 months required for the risk estimate, so no historical proxies are used.[14]

Our active risk-free series is the team's FRED DGS1MO download: the annualized one-month Treasury constant-maturity yield, in percent. We average the available daily observations within each month and divide by 1,200 to obtain a monthly yield proxy. We use that proxy in excess returns and Sharpe calculations. The professor's series is comparison evidence only. This conversion is an approximation to a monthly risk-free return, rather than a Treasury-bill total-return index.[9][14]

Table 4. Annualized volatility over the latest 60 complete months

ETF	Annual volatility	Historical excess mean	Estimation period
VOO	15.67%	10.34%	Oct 2021–Sep 2026
VTV	13.99%	8.91%	Oct 2021–Sep 2026
IEFA	15.71%	6.03%	Oct 2021–Sep 2026
EEM	17.68%	5.71%	Oct 2021–Sep 2026
IEF	7.65%	-5.62%	Oct 2021–Sep 2026
IGSB	3.36%	-1.55%	Oct 2021–Sep 2026
GLD	17.20%	14.55%	Oct 2021–Sep 2026
DBC	17.28%	9.87%	Oct 2021–Sep 2026

Source: BlackHawk Capital, own calculations and compilation. [14]

Table 4 shows the inputs used for formation. The covariance matrix comes from the same October 2021-September 2026 window and is annualized by multiplying by twelve. Individual standard deviations are annualized by the square root of twelve. The historical excess means are included for transparency: the chosen risk-contribution method does not use them to set the weights.[14]

3.2 Return correlation and economic diversification

The four equity funds share a substantial market component. International stocks and a value tilt broaden the businesses and regions we own, but they can still fall together. Figure 4 measures this co-movement from monthly returns over the formation window.[14]

This matrix is where the asset universe becomes a portfolio. An ETF's risk depends on its own volatility and on how it moves with the other holdings. Treasury duration, corporate credit, bullion and commodity futures supply different economic exposures, although their diversifying effect varies with the regime. We therefore use the complete covariance matrix when assigning risk contributions.[14]

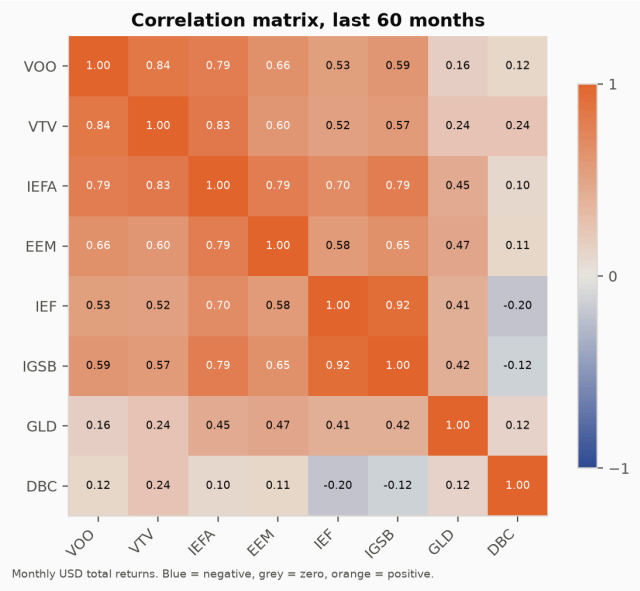


Figure 4. Monthly return correlations over the final 60 months

Source: BlackHawk Capital, own calculations and compilation. [14] Monthly return correlations, October 2021-September 2026.

4 Allocation methodology

4.1 The decision we ask investors to support

We choose Capped Equal Risk Contribution because it starts with the risks we are willing to hold. The method gives each ETF a risk-budget target and accounts for correlations when converting those targets into capital weights. It suits an allocation built from broad asset classes whose future returns are difficult to estimate precisely.[14]

Our decision has three layers. First, we choose the exposures: geography and value within equities, Treasury duration and short credit within bonds, and separate gold and commodity sleeves. Second, we put limits on concentrations we are unwilling to accept. Third, the risk-contribution rule translates those choices into weights.[14]

4.2 From monthly returns to a portfolio risk estimate

The starting point is the last 60 complete monthly USD returns. For each ETF, the sample variance measures its variability; each pairwise covariance measures their joint movement. We annualize this matrix, denoted Sigma, and calculate the portfolio's annual volatility from the proposed weights.[14]

$$\Sigma = 12 \times \mathrm{Cov}(\text{monthly returns}) \qquad \sigma_p = \sqrt{(w'\Sigma w)}\,[14]$$

Here w is the vector of capital weights. The cross terms in this calculation explain why we cannot obtain portfolio volatility by simply averaging the eight individual volatilities. A position can diversify the portfolio even if its own volatility is high, provided its co-movement with the other positions is sufficiently different.[14]

For ETF i, the notebook calculates its normalized risk contribution as its weight multiplied by its covariance with the whole portfolio, divided by total portfolio variance. The resulting shares sum to 100% and are also the normalized contributions to portfolio volatility.[14]

$$RC_i = w_i(\Sigma w)_i / (w'\Sigma w) \qquad \Sigma_i\, RC_i = 1\,[14]$$

This measure answers the question we care about: how much of the portfolio's total risk comes from each position? It allows us to distinguish a large capital allocation to a relatively quiet asset from a smaller allocation that drives a large share of portfolio risk.[14]

4.3 The risk-budget objective and our concentration limits

With eight funds, the initial target is 12.5% of total risk per ETF. The optimizer minimizes the sum of squared differences between each actual risk share and this target. It searches across non-negative weights that sum to one, subject to the IGSB, GLD and DBC caps. Equal risk shares are the objective; equal capital weights are not.[14]

$$\text{Minimize } \Sigma_i \, (RC_i - 1/8)^2 \quad \text{subject to } w_i \geq 0 \text{ and } \Sigma_i \, w_i = 1$$[14]

The 12% IGSB cap is central to our decision. Inverse-volatility weighting and uncapped Equal Risk Contribution allocate roughly one third of capital to this single short corporate-bond ETF. We want short credit in the portfolio, but we also want the risk reduction to come from several exposures.[14]

Our estimates are uncertain, and a risk-budgeting optimizer treats the supplied covariance matrix as known. Michaud's discussion concerns mean-variance optimization; we use it as a warning about dependence on estimated inputs, rather than as proof that our capped ERC problem has the same error amplification. The cap limits the capital consequence of an estimate we are unwilling to rely on without reservation.[21][14]

IGSB's estimated annual volatility is 3.36% over the formation window. Its short duration provides an economic reason for relatively low rate sensitivity. Our full-history realized estimate is lower, about 2.52%, so the data do not establish that the recent rate shock artificially compressed its risk. We cap the position because uncapped rules put roughly one third of capital in this single credit ETF, while future volatilities and correlations can change. DeMiguel, Garlappi and Uppal illustrate the difficulty of estimation in a different setting: their US-equity calibration requires around 3,000 months for a sample-based mean-variance portfolio with 25 assets to outperform equal weighting. That figure is context for uncertainty, not a required sample length or a validation result for our eight-ETF ERC allocation.[22][11][14]

The 15% limits on GLD and DBC serve a different purpose. Gold is exposed to real yields, the dollar and investor demand; broad commodity futures are exposed to supply conditions, the business cycle and roll effects. Their diversifying role does not remove these risks. Separate caps limit how much capital either exposure can absorb. Our 50% equity ceiling preserves a meaningful allocation to the other sleeves.[14]

The code solves this capped risk-contribution problem first. It then checks that the chosen weights are fully invested and long-only, that equity is at or below 50%, and that formation volatility is between 7% and 9%. The volatility band and equity ceiling are acceptance checks for this allocation; they are not included as extra constraints in the capped ERC solver. The resulting 8.58% passes those checks. We retain it within the permitted band rather than forcing an exact 8% through leverage or a cash adjustment.[14]

These limits are discretionary and were chosen while reviewing the historical evidence. They express the structure we prefer to manage, and we disclose that selection when interpreting the backtests. They do not establish that the chosen weights maximize the challenge's future Sharpe Ratio.[14]

4.4 What the resulting weights mean

The solution places 37.21% of capital in equities, 36.04% in bonds, and 26.75% in gold and broad commodities. The four equity funds together contribute about 56.09% of estimated risk. This is why a diversified capital allocation can still be strongly influenced by an equity shock.[14]

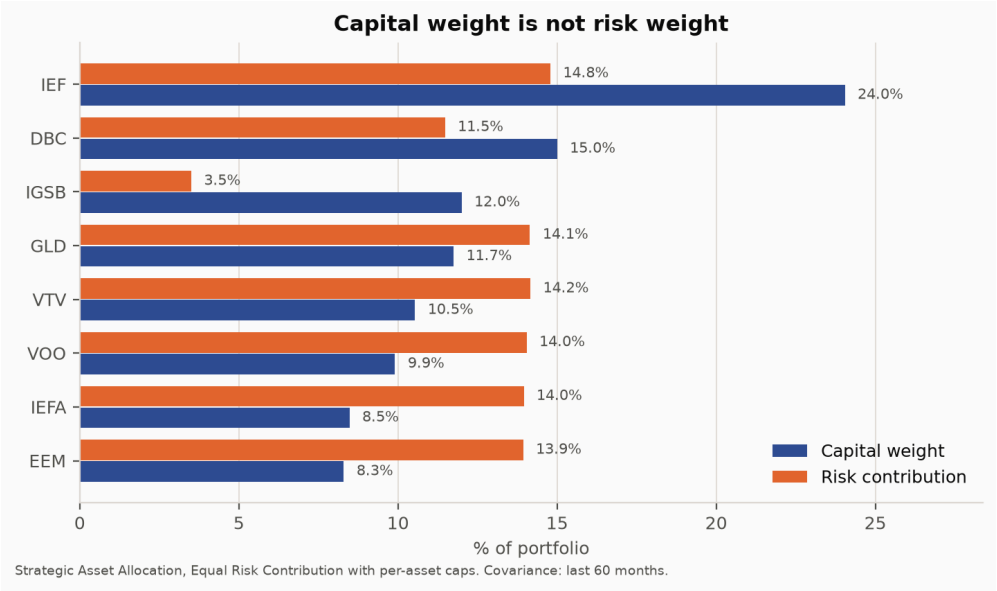


Figure 5. Capital allocation versus normalized risk contribution

Source: BlackHawk Capital, own calculations and compilation. [14] Capital weights and normalized risk contributions; final 60-month sample covariance.

IEF is the largest capital position at 24.04%, with a risk share of 14.79%. IGSB reaches its 12% capital cap and contributes only 3.50% of risk. DBC reaches its 15% cap and contributes 11.48%; GLD remains below its cap at 11.75% of capital. The other six risk contributions are approximately 14-15% each. The binding limits explain why the final portfolio does not have eight exactly equal risk shares.[14]

The higher Treasury weight follows from the covariance estimate and the risk-budget objective. The weights were not manually set to a conventional equity/bond ratio. Similarly, the real-asset allocation is the result of the chosen universe, covariances and limits; it is not an extrapolation of DBC's recent return.[14]

We translate the 8.58% annual volatility into a one-month 5% normal VaR using the zero-mean convention applied in class. The calculation gives 4.07% of capital, or USD 4.07 million. It is useful for communicating the scale of risk, while the historical drawdowns and stress tests show how losses can develop beyond a single normal-distribution estimate.[14]

$$\text{One-month 5\% VaR} = 1.645 \times \sigma_p / \sqrt{12} = 4.07\% \text{ of capital}[14]$$

4.5 Why we chose this method over the alternatives

We compared Equal Weight, inverse-volatility Risk Parity, uncapped Equal Risk Contribution, our capped version and constrained Maximum Sharpe. We also calculated a capped Minimum Variance diagnostic. Table 5 separates current formation estimates from the Sharpe realized by the historical walk-forward method.[14]

Table 5. Allocation methods and the reason for our choice

Method	Formation vol	Fitted Sharpe	Walk-forward Sharpe	Assessment
Equal Weight	9.66%	0.625	0.661	9.66% exceeds the risk ceiling
Inverse-volatility Risk Parity	7.26%	0.383	0.587	37.2% in IGSB exceeds our cap
Equal Risk Contribution	7.06%	0.469	0.718	34.1% in IGSB exceeds our cap

Source: BlackHawk Capital, own calculations and compilation. [14]

Table 5. Allocation methods and the reason for our choice (continued)

Method	Formation vol	Fitted Sharpe	Walk-forward Sharpe	Assessment
Capped Equal Risk Contribution	8.58%	0.537	0.583	Selected: risk band and instrument caps
Constrained Maximum Sharpe	9.00%	0.987	0.691	Uses fitted excess-return estimates
Capped Minimum Variance	6.02%	-0.467	-	6.02% falls below the risk floor

Source: BlackHawk Capital, own calculations and compilation. [14]

Equal Weight is attractive for its simplicity, but its formation volatility is above 9%. Inverse-volatility Risk Parity brings volatility inside the band by giving more capital to low-volatility bonds. Its use of individual volatilities overlooks differences in correlations. Full Equal Risk Contribution uses those correlations, but the uncapped solution still puts 34.1% in IGSB. We prefer to accept unequal risk shares within our caps.[14]

Constrained Maximum Sharpe uses the latest 60-month historical excess means as return estimates, alongside covariance, a 25% instrument bound and the 7-9% risk band. Its fitted Sharpe is higher, but that result relies on the return estimates we are least confident in. Its historical walk-forward Sharpe is also above ours; we acknowledge that evidence while preferring a rule whose weights do not depend on forecasting those means.[14]

Capped Minimum Variance reaches 6.02% formation volatility, below the 7% floor, so that particular diagnostic is ineligible. It was solved without the floor; a separately constrained problem could give a different answer. We also plotted a Maximum Sharpe diagnostic without the volatility band, which reaches 10.78% and exceeds the ceiling. Neither diagnostic changes our selected allocation.[14]

This is a choice of process and portfolio structure. We are not selecting the highest Sharpe observed among every historical experiment. The blend analysis in Section 7 shows the cost of this preference directly, and the covariance and window tests show how dependent the weights are on the risk estimates. [14]

We read the differences between backtested Sharpe ratios against their sampling uncertainty. Table 6 uses the normal-theory IID approximation SE(SR annual) = sqrt(12) × sqrt((1 + SR monthly squared / 2) / T), with T = 107 monthly observations. This formula additionally relies on normal-return and effectively fixed risk-free assumptions; our varying monthly yield proxy makes it an approximation to the reported Sharpe convention. The standard errors are about 0.34 in annualized units, and the approximate 95% interval for our gross method is -0.08 to 1.24. These intervals describe sampling uncertainty under the assumptions; they do not correct for research selection.[17][14]

Table 6. Precision of the walk-forward Sharpe ratios

Allocation	Sharpe	Std. error	95% interval	Gap vs ours	Paired p	Bootstrap p
Capped ERC (our SAA)	0.583	0.337	-0.08 to 1.24	-	-	-
Equal Weight	0.661	0.338	-0.00 to 1.32	+0.077	0.52	0.61
Inverse-volatility Risk Parity	0.587	0.337	-0.07 to 1.25	+0.003	0.96	0.97
Equal Risk Contribution	0.718	0.338	+0.05 to 1.38	+0.135	0.12	0.16
Constrained Maximum Sharpe	0.691	0.338	+0.03 to 1.35	+0.107	0.48	0.39

Source: BlackHawk Capital, own calculations and compilation. [17][18][19][20][14]

The allocation returns are 90-98% correlated, so paired comparisons account for their shared variation. We report the Jobson-Korkie test with the Memmel correction as a normal-theory diagnostic, alongside a separate circular block-bootstrap sensitivity check using six-month blocks and 10,000 paired resamples.

This non-studentized bootstrap preserves local dependence and relies on suitable stationarity, weak-dependence and block-length assumptions; it is not the exact studentized procedure proposed by Ledoit and Wolf. None of the four comparisons against capped ERC resolves an advantage at the 5% level. Uncapped ERC has the largest gap, 0.135, with p = 0.12 in the normal-theory test and p = 0.16 in this bootstrap. USMV is not included in these paired tests, so its addition-test gain has no significance conclusion from this table.[18][19][20][14]

These tests leave insufficient evidence that any of the four alternatives outperforms our selected rule. They do not establish equality of the five methods or test every pair among them. We therefore give weight to the risk budget and portfolio structure we want to manage, alongside the uncertain historical rankings. The halving discount in Section 6.1 separately addresses research selection. Performance over the short live challenge will provide another, particularly noisy observation of this process.[17][13][14]

4.6 Comparison with the robust rules (Checkpoint 2)

Equal Weight assigns 12.5% to every ETF, yet the equity funds carry disproportionate risk; its formation volatility is 9.66%, above the AMC ceiling. Inverse-volatility Risk Parity reduces that estimate to 7.26%, but places 37.2% of capital in IGSB, against our 12% credit-instrument cap. Our allocation reaches 8.58% with a more balanced capital structure. We choose it for that structure and for its use of correlations, while acknowledging the historical Sharpe advantages of other rules.[14]

Table 7. Checkpoint 2 weights and risk contributions

ETF	Our SAA w / RC	Equal Weight w / RC	Risk Parity w / RC
VOO	9.9% / 14.0%	12.5% / 16.4%	8.0% / 13.9%
VTV	10.5% / 14.2%	12.5% / 15.2%	8.9% / 14.3%
IEFA	8.5% / 14.0%	12.5% / 18.7%	8.0% / 16.1%
EEM	8.3% / 13.9%	12.5% / 19.4%	7.1% / 14.4%
IEF	24.0% / 14.8%	12.5% / 6.6%	16.3% / 13.2%
IGSB	12.0% / 3.5%	12.5% / 3.2%	37.2% / 14.3%
GLD	11.8% / 14.1%	12.5% / 13.0%	7.3% / 9.7%
DBC	15.0% / 11.5%	12.5% / 7.6%	7.2% / 4.0%
Ex ante vol	8.58%	9.66%	7.26%

Source: BlackHawk Capital, own calculations and compilation. [14]

5 Portfolio diversification

5.1 Asset classes and geographic exposure

Our fund-level mix spreads capital across equities, two types of bonds and two real-asset exposures. Within equities, the holdings data show a more precise picture than the listing currency or the name of the ETF.[14]

US-classified stocks account for 53.16% of the equity sleeve, equivalent to 19.78% of the whole fund. Including IEF and IGSB brings US equity-plus-bond exposure to approximately 55.82% of total capital under these classifications. We are comfortable retaining a substantial US allocation, but international equity positions do not make the fund geographically neutral.[14]

Taiwan, Japan, South Korea and China represent about 6.44%, 6.06%, 4.76% and 3.96% of the equity sleeve. These are the provider's country groupings at the common September snapshot. The exports do not document the attribution rule or measure revenue geography. USD listing provides a common reporting currency; it does not hedge the currencies of the underlying international holdings.[14]

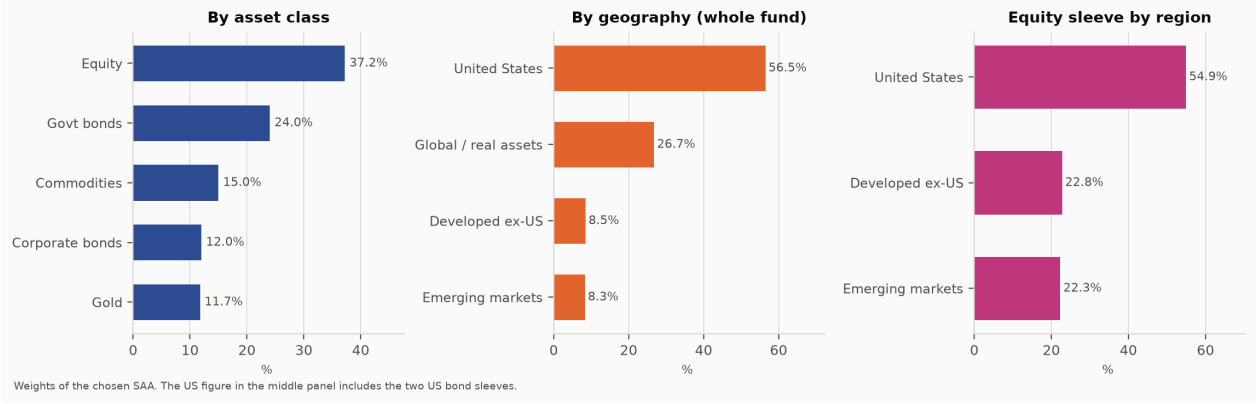


Figure 6. Allocation by asset class and geographic mandate

Source: BlackHawk Capital calculations on FactSet data. [12][14] Fund mandates and asset-class groupings; global real assets shown separately.

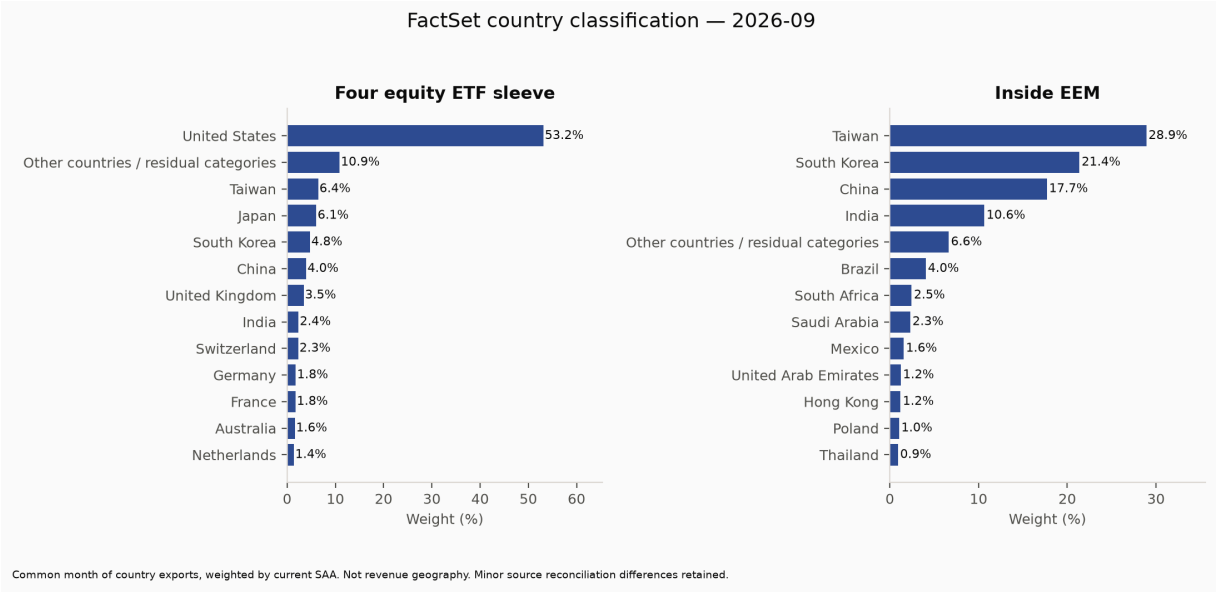


Figure 7. Stock-country composition within the equity sleeve

Source: BlackHawk Capital calculations on FactSet data. [12][14] Common September 2026 provider country snapshot; revenue geography is not measured.

5.2 Equity sectors and the value allocation

The value sleeve changes what we own inside the US allocation. It reduces some of the technology emphasis and increases exposure to other sectors, including financial companies. That makes the style decision visible in the holdings, rather than relying only on the ETF label.[14]

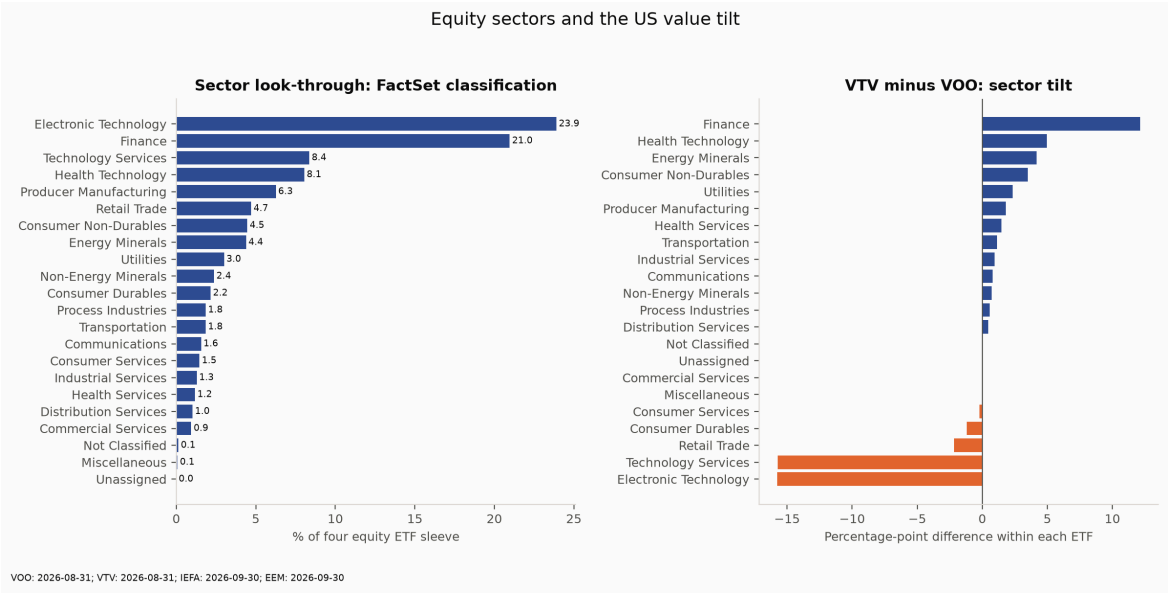


Figure 8. Sector composition and VTV relative to VOO

Source: BlackHawk Capital calculations on FactSet data. [12][14] Reported sector categories and VTV versus VOO; mixed August and September snapshots.

Electronic Technology accounts for 23.90% of the equity sleeve and Technology Services for a further 8.37%. Technology therefore remains a meaningful risk even after the value tilt and geographic diversification. Sector snapshots are descriptive, dated evidence: US fund data are from August and international fund data from September, and the classifications follow the provider's sector system.[14]

5.3 Security overlap

VOO and VTV have 41.91% weighted security overlap after normalizing identified equities. We accept this overlap because VTV is a deliberate style tilt within the US sleeve. It changes the balance of the same broad opportunity set rather than adding a wholly independent asset class.[14]

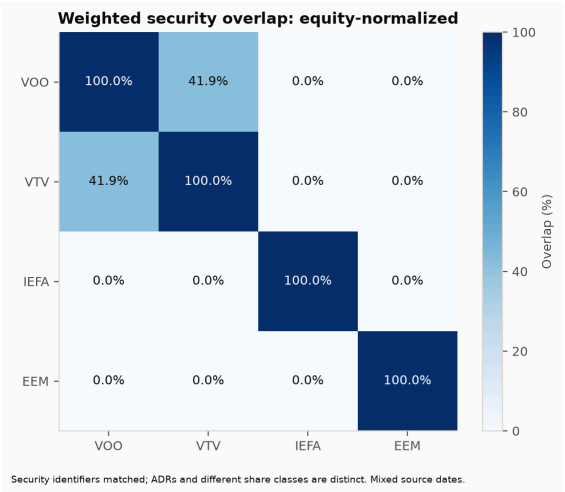


Figure 9. Weighted overlap between the equity ETFs

Source: BlackHawk Capital calculations on FactSet data. [12][14] Matched security identifiers, normalized within identified equities.

The overlap measure matches security identifiers and keeps share classes, ADRs and local lines separate. It answers an ownership question. Correlation answers a return question: two funds can own different securities and still respond to the same macro shock.[14]

5.4 Largest stock positions

The largest combined stock position is Taiwan Semiconductor, at about 1.32% of total capital through EEM. The position is modest at fund level, while the concentration within EEM is more substantial. Its ten largest stocks account for 38.80% of that ETF; VOO's top ten account for 39.09%, compared with 23.58% in VTV and 12.16% in IEFA.[14]

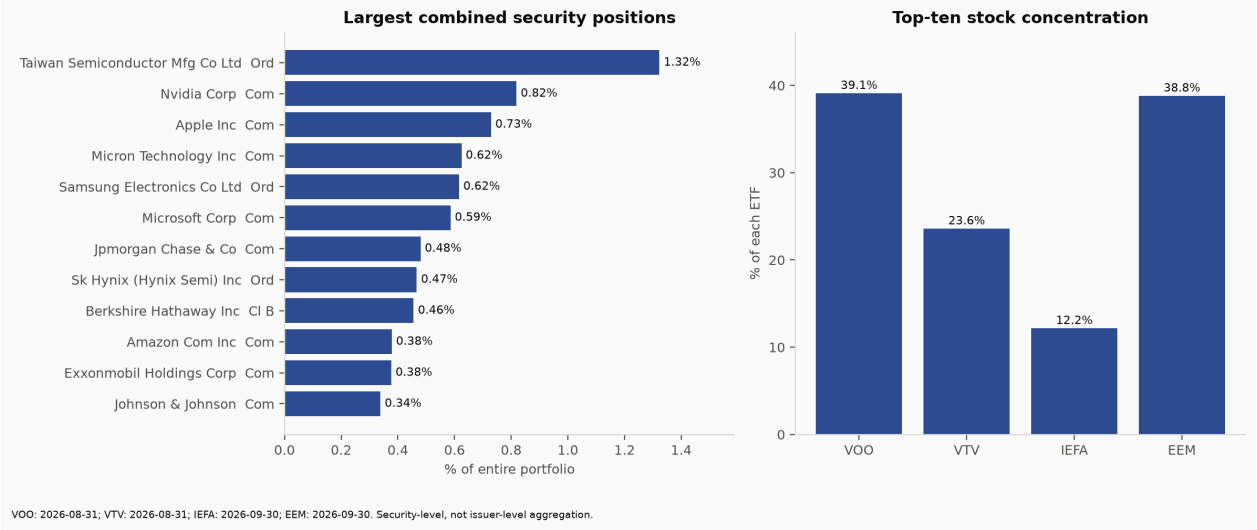


Figure 10. Largest combined security positions and ETF concentration

Source: BlackHawk Capital calculations on FactSet data. [12][14] Combined stock positions at the proposed weights; distinct share classes remain distinct.

These holdings qualify our diversification claim. We own eight broad funds, yet parts of the equity book depend on a relatively small group of large businesses. The portfolio weights reduce the effect of any one stock on the whole fund; they do not eliminate technology, regional or market-wide risk.[14]

6 Backtest and historical risk

6.1 Chronological backtest protocol (Checkpoint 3)

Design: the universe is the eight ETFs of Table 3; the rule that determines the weights is Capped Equal Risk Contribution; the portfolio is rebalanced monthly; and the backtest covers November 2012 to September 2026, with the walk-forward test beginning in November 2017 once the first 60-month estimation window is complete. In the chronological test, each month's weights use only the previous 60 complete monthly returns, then earn the following month's return. Risk-free observations are aligned to the month in which returns are realized. Refitting the method monthly lets us assess how the allocation process behaved as its information changed.[14]

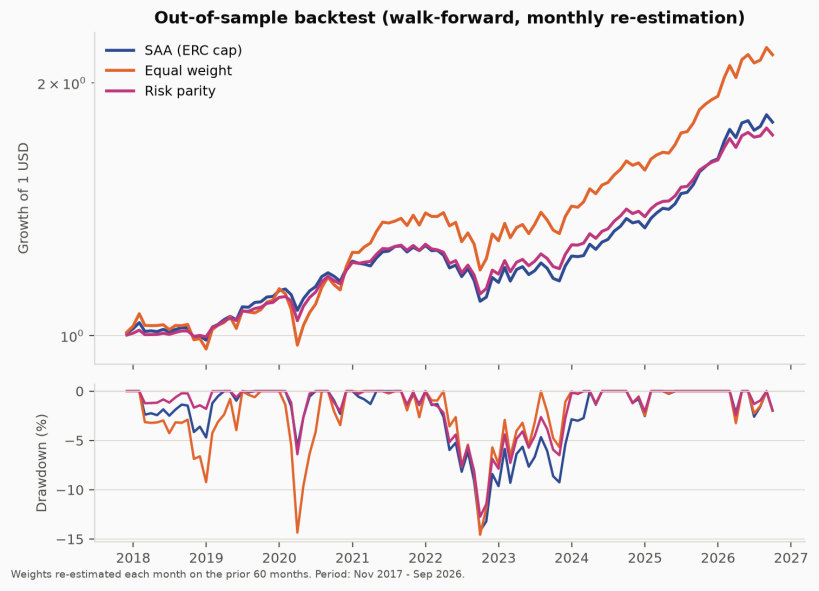


Figure 11. Walk forward cumulative performance and diagnostic comparison

Source: BlackHawk Capital, own calculations and compilation. [14] November 2017-September 2026; monthly fits using only the preceding 60 months.

This historical re-estimation differs from the live AMC portfolio, whose positions remain unchanged between rounds. It also does not remove the effect of choosing the ETF universe and concentration limits after reviewing history. The project therefore has no untouched external holdout for all research choices.[14]

We document 34 economic configurations and protocol attempts: 32 in the original register, plus the relaxed-output-cap blend and the separate Maximum Sharpe diagnostic without a volatility band. Appendix C records them, including unsuccessful earlier trials. This is a minimum documented research count, rather than a count of numerical optimizer iterations.[14]

The blend experiment has its own chronological split: training through September 2023 and a frozen-weight test from October 2023 to September 2026. We describe that protocol separately because its component weights and method coefficients are held fixed during the test.[14]

Realistic discounted Sharpe: 0.29. The net walk-forward Sharpe of 0.579 is the historical starting point and we halve it, following the discount rule given in Class 4 as a rule of thumb, with larger discounts when there are more tunable parameters. Three features of this project justify the full halving rather than a lighter haircut: we document 34 configurations and protocol attempts; the ETF universe and the instrument caps were chosen after reviewing history, so no untouched external holdout exists for all research choices; and the common sample begins in November 2012, after the 2008 crisis, so it omits that financial crisis. We therefore carry 0.29 as the Sharpe we consider realistic going forward, and treat a higher realized outcome as favourable rather than as the expected case.[14][23]

6.2 Performance measurement

The historical evidence supports a moderate-risk allocation, with meaningful losses along the way. Table 8 distinguishes the retrospective performance of today's fixed weights from the performance of the risk-allocation method when refitted through time.[14]

Table 8. Three distinct historical evaluations

Measure	Current fixed SAA	Walk forward gross	Walk forward net
Ann. return	6.12%	6.84%	6.81%
Ann. excess return	4.34%	4.18%	4.15%
Ann. vol	7.54%	7.17%	7.17%
Sharpe	0.576	0.583	0.579
t-stat (excess)	2.16	1.76	1.74
Max drawdown	-13.10%	-14.17%	-14.19%
VaR 5% 1m	3.58%	3.40%	3.40%

Source: BlackHawk Capital, own calculations and compilation. [14]

Class 4 asks us to report the backtested mean with its t-statistic. Table 8 pairs annualized excess means of 4.34%, 4.18% and 4.15% with excess-mean t-statistics of 2.16, 1.76 and 1.74. We calculate t as mean monthly excess return divided by its standard error, sd(monthly excess return) / sqrt(T). The fixed allocation exceeds the approximate 1.96 two-sided threshold; the gross and net walk-forward results do not. These are unadjusted IID diagnostics, with no serial-dependence or data-mining correction. For comparison, the fixed allocation's total-return t-statistic is 3.03. The identity t = monthly Sharpe × sqrt(T) is exact when Sharpe uses the same excess-return standard deviation. Our reported Sharpe uses total-return volatility and the risk-free proxy varies, so that identity is only approximate here.[23][14]

The fixed-weight simulation covers 167 months, November 2012-September 2026, with monthly rebalancing. Annualized arithmetic return is 6.12%, realized volatility 7.54%, Sharpe 0.576 and maximum drawdown 13.10%. Arithmetic annualization is twelve times the mean monthly return; it differs from a compound annual growth rate.[14]

The walk-forward test starts in November 2017, after the initial 60-month fit. Gross Sharpe is 0.583; charging the stated trading-cost assumption reduces it to 0.579. Net realized volatility is 7.17% and net maximum drawdown is 14.19%. The different dates and changing weights mean this is a separate test from the fixed-allocation simulation.[14]

Table 9 reports the Class 2 measures for the fixed allocation. Round 1 has no common challenge benchmark, so we declare VOO as the equity reference for CAPM quantities and a monthly-rebalanced 60/40 mix of VOO and IEF as a multi-asset diagnostic. Sortino follows the course convention: mean excess return divided by downside deviation below MAR = 0. In the full 167-month sample, our Sharpe of 0.576 is below the 60/40 at 0.840 and VOO at 0.936; alpha against VOO is -1.19%. VOO's 13.20% annual excess return is an arithmetic annualized mean, not a compound growth rate. The ranking reflects the exposures and period observed; a lower volatility target alone does not mechanically imply a lower Sharpe Ratio. Information Ratio in this table is CAPM alpha divided by residual volatility; the tracking-error version is retained separately in the code outputs.[14][24]

Table 9. Class 2 performance measures: full sample and formation window

Measure	SAA 167m	60/40 167m	VOO 167m	SAA 60m	60/40 60m	VOO 60m
Ann. excess return	4.34%	7.63%	13.20%	4.60%	3.96%	10.34%
Ann. volatility	7.54%	9.08%	14.10%	8.58%	11.34%	15.67%
Sharpe	0.576	0.840	0.936	0.537	0.349	0.660
Sortino (MAR 0)	0.931	1.382	1.554	0.879	0.543	1.108
Treynor	0.104	0.123	0.132	0.109	0.056	0.103
Beta vs VOO	0.419	0.619	1.000	0.421	0.702	1.000
Alpha vs VOO	-1.19%	-0.54%	-	0.24%	-3.30%	-
IR (alpha / residual vol)	-0.256	-0.217	-	0.045	-1.286	-
Upside capture	43.2%	62.8%	100%	51.4%	66.8%	100%
Downside capture	45.9%	62.8%	100%	44.1%	77.6%	100%
Max drawdown	-13.10%	-20.50%	-23.91%	-13.10%	-20.50%	-23.91%
Skewness	-0.37	-0.37	-0.39	-0.48	-0.38	-0.28
Excess kurtosis	0.73	0.77	0.87	-0.02	-0.16	-0.26

Source: BlackHawk Capital, own calculations and compilation. [14]

What the allocation does deliver over the full sample is loss control: a 13.10% maximum drawdown against 20.50% for the 60/40 and 23.91% for VOO, and the lowest downside deviation of the three. In the 60-month formation window the ranking changes, with our Sharpe of 0.537 above the 60/40 at 0.349, Treynor of 0.109 above VOO at 0.103, alpha positive at 0.24% and upside capture of 51.4% above downside capture of 44.1%. That window is the one used to estimate the covariance, so it is in-sample for the weighting rule and we do not present it as forward evidence. Capture ratios are measured on the months when VOO rises or falls; the full-sample pair, 43.2% up against 45.9% down, is mildly unfavourable and we report it rather than omit it.[14]

The VaR rows above use each backtest's realized volatility. Current formation VaR instead uses the final 60-month covariance matrix and equals 4.07%. For the fixed allocation, historical monthly 5% VaR is 3.03%, historical Expected Shortfall is 4.23%, and the worst month lost 7.50%.[14]

Drawdown is the decline from a previous wealth peak. Our monthly observations can miss deeper losses within a month, and the common sample starts after the 2008 crisis. We account for that limited coverage when deciding how much confidence to place in the historical results.[14]

6.3 Fixed allocation simulation

The fixed-weight backtest answers a practical question: how would the allocation we propose today have behaved over the common history? We rebalance monthly to the current weights and reinvest the return series. The cumulative wealth and drawdown paths in Figure 12 show both the compounding and the losses investors would have had to withstand.[14]

Because today's weights were estimated at the end of the sample, this simulation is retrospective. It is useful for describing the allocation's behavior, while the chronological test below is better suited to assessing the repeated weighting process. The equal-weight and inverse-volatility series are diagnostic comparators; the challenge has no common benchmark.[14]

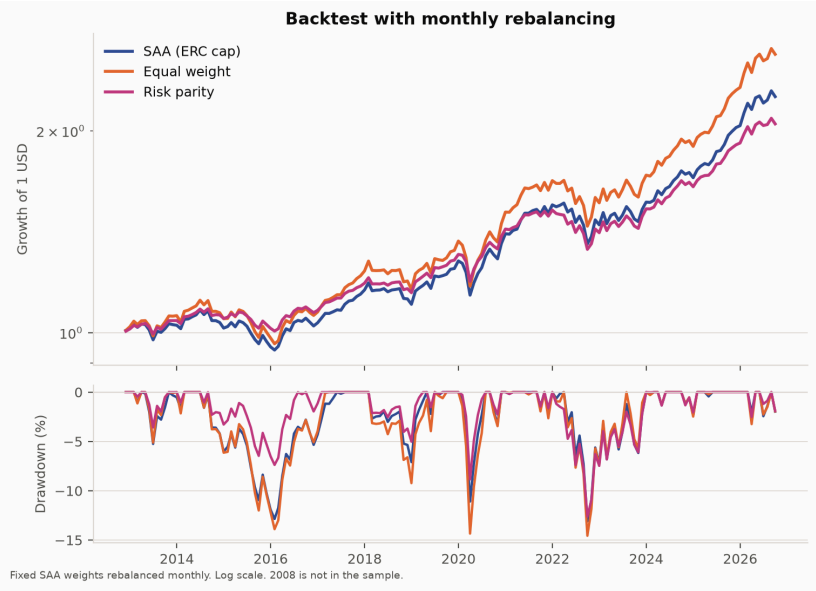


Figure 12. Cumulative wealth and drawdown of today's allocation

Source: BlackHawk Capital, own calculations and compilation. [14] November 2012-September 2026; monthly rebalancing to current fixed weights.

6.4 Performance across time

The recent experience is stronger than the full-sample result. With today's fixed weights, the three-year Sharpe is 1.50, compared with 0.576 across the common history. Figure 13 shows how much that measure and realized volatility change across 36-month windows.[14]

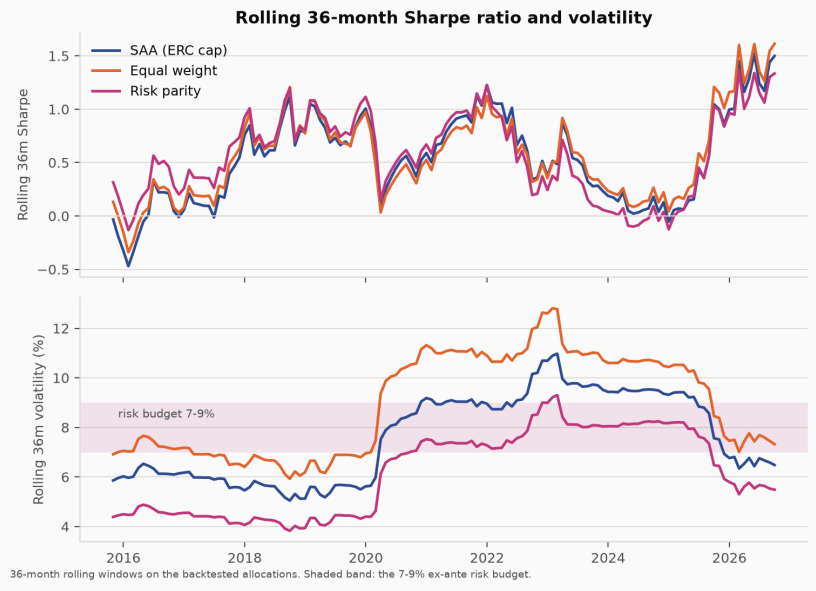


Figure 13. Rolling volatility and Sharpe over 36 months

Source: BlackHawk Capital, own calculations and compilation. [14] Rolling 36-month realized estimates; the 7-9% band is a formation requirement.

Table 10. Current fixed allocation across historical horizons

Horizon	Arithmetic return	Volatility	Sharpe	Max drawdown
1 year	13.67%	8.28%	1.191	-2.44%
3 years	14.29%	6.48%	1.499	-2.44%
5 years	8.43%	8.58%	0.537	-13.10%
Full sample	6.12%	7.54%	0.576	-13.10%

Source: BlackHawk Capital, own calculations and compilation. [14]

6.5 Stress windows and tail risk

In the selected historical stress windows, diversification reduced losses relative to the equity comparators. The current SAA lost 10.35% from February to March 2020, against 19.55% for VOO. From January to October 2022 it lost 10.34%, against 17.74% for VOO. Investors still had to absorb material losses in both episodes.[14]

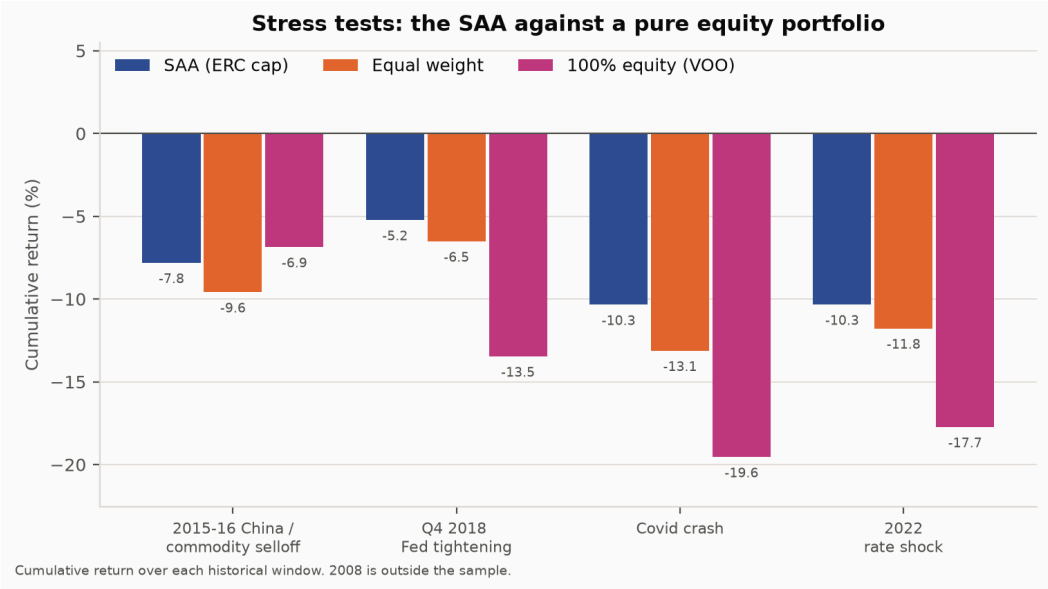


Figure 14. Returns during selected historical stress periods

Source: BlackHawk Capital, own calculations and compilation. [14] Selected historical periods at current fixed weights; returns are not maximum drawdowns.

Table 11. Hypothetical one month shocks applied to current weights

Scenario	Portfolio return	Loss USD m
Equity crash	-7.72%	7.72
Inflation shock	-1.66%	1.66
Rate shock (2022-like)	-6.37%	6.37

Source: BlackHawk Capital, own calculations and compilation. [14]

Table 12. Assumed one month returns behind each hypothetical scenario

Scenario	VOO	VTV	IEFA	EEM	IEF	IGSB	GLD	DBC
Equity crash	-20%	-18%	-20%	-25%	3%	1%	5%	-10%
Inflation shock	-10%	-8%	-10%	-12%	-8%	-2%	10%	20%
Rate shock (2022-like)	-15%	-12%	-15%	-15%	-10%	-3%	-5%	15%

Source: BlackHawk Capital, own calculations and compilation. [14]

Table 12 gives the assumed returns behind each scenario, so that the aggregate figures can be checked against our weights. They are judgement-based assumptions, not estimates from a model, and their probabilities are unspecified. A scenario combining equity and bond weakness is particularly relevant to the nominal-duration exposure, while a commodity reversal would hurt the real-asset sleeve.[14]

7 Alternative allocations and robustness

7.1 Additions to the universe

We tested whether an extra position improved the eight-fund structure. TIP, LQD, EMB, USMV and VNQ were each added separately, with a 12% candidate cap and the same capped ERC tools. For each experiment we aligned the histories, recalculated correlations and weights, and evaluated formation risk and historical performance.[14]

Table 13. One additional ETF at a time

Candidate	Highest correlation	Form. vol	Fixed Sh.	Walk-fwd Sh.	Change	Max DD	Band	Verdict
8 ETF (current SAA)	-	8.58%	0.576	0.583	+0.000	-13.10%	yes	baseline
+ TIP	IEF 0.87	7.97%	0.524	0.543	-0.040	-12.58%	yes	reject
+ LQD	IGSB 0.93	8.37%	0.526	0.520	-0.064	-12.94%	yes	reject
+ EMB	IEFA 0.90	8.45%	0.516	0.515	-0.068	-13.34%	yes	reject
+ USMV	VTV 0.89	8.56%	0.617	0.591	+0.007	-12.94%	yes	reject
+ VNQ	VTV 0.84	8.95%	0.558	0.537	-0.046	-13.74%	yes	reject

Source: BlackHawk Capital, own calculations and compilation. [14]

USMV raises gross walk-forward Sharpe from 0.583 to 0.591, an improvement of 0.0075; the other individual additions lower it under this protocol. We judge that small gain insufficient to justify another position in the current structure. This decision leaves room to revisit an alternative if its economic role or implementation case becomes more compelling.[14]

The +0.03 Sharpe threshold in the research register was introduced after inspecting the results. It is a screening choice, not evidence of statistical significance. Correlation helps explain the overlap in behavior, but the decision also considers the allocation change, concentration and number of positions. [14]

7.2 Substitutions within sleeves

Substitution tests address a different question: can another ETF perform the same strategic role more effectively? We replace one fund at a time, use the aligned common months, then recompute the capped weights and evaluate the result with the same tools.[14]

Table 14. Substitution results

Replacement	Formation vol	Walk forward Sharpe	Change	In band
VOO -> SPY	8.58%	0.583	-0.001	yes
IEFA -> EFA	8.56%	0.584	+0.001	yes
EEM -> VWO	8.44%	0.580	-0.004	yes
VTV -> USMV	8.41%	0.572	-0.011	yes
IEF -> TLT	9.94%	0.548	-0.035	NO
IGSB -> LQD	9.27%	0.539	-0.045	NO
IGSB -> TIP	8.86%	0.566	-0.017	yes
IGSB -> EMB	9.35%	0.530	-0.053	NO
GLD -> IAU	8.58%	0.587	+0.003	yes

Source: BlackHawk Capital, own calculations and compilation. [14]

EFA and IAU produce small positive walk-forward Sharpe differences; SPY and VWO are close to the baseline. These gaps are too small to change our conviction in the structure. The tested TLT, LQD and EMB substitutions exceed 9% formation volatility, which explains their rejection under those particular weights. Different weights could make the same funds admissible.[14]

Replacing IGSB with TIP changes the economic exposure from short corporate credit to inflation-linked duration, so the choice involves more than comparing Sharpe ratios. We retain short credit in this round. Broad commodities have no substitution test in this study.[14]

7.3 Blending allocation methods

We also asked whether a combination of weighting rules could improve on a single rule. The blend uses six components: Equal Weight, inverse-volatility Risk Parity, uncapped Equal Risk Contribution, Capped Equal Risk Contribution, constrained Maximum Sharpe and capped Minimum Variance. Non-negative method coefficients sum to one, and each coefficient determines how much of that component's ETF weights enters the final portfolio.[14]

For the training fit, each component uses the last 60 months available at September 2023. The blend coefficients maximize historical training Sharpe using the full training history, with formation volatility checked against the final 60-month training covariance. The numerical search starts from an equal method mix and from each pure method, then selects the best successful feasible solution found. Multiple starting points reduce dependence on one initial guess; the result remains specific to these component portfolios and numerical fits.[14]

We freeze both component ETF weights and method coefficients for the October 2023-September 2026 test. A separate latest-data fit checks the current capped blend, but does not determine the held-out returns. These frozen weights implement a constant-weight, monthly-rebalanced simulation; they are not the changing walk-forward weights.[14]

With our instrument caps, the fitted blend selects effectively 100% Capped Equal Risk Contribution. Its training and holdout results therefore coincide with that component. Removing the output instrument caps produces a different mix, while retaining the 50% team equity ceiling, the 7-9% formation band and the original component construction rules.[14]

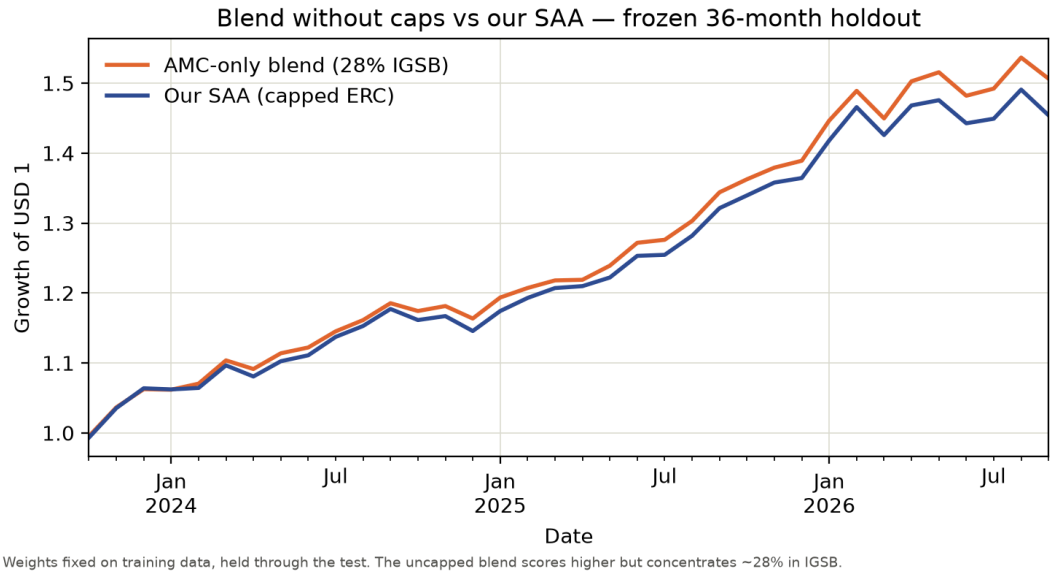


Figure 15. Blend after removing the project instrument caps

Source: BlackHawk Capital, own calculations and compilation. [14] October 2023-September 2026; training-fitted weights held constant with monthly rebalancing.

7.4 The concentration trade off

The less constrained blend has a higher historical test Sharpe: 1.537 against 1.286 for capped ERC fitted at the same training date. It combines 53.58% inverse-volatility Risk Parity, 31.49% Equal Weight and 14.93% Maximum Sharpe; the other method coefficients are zero. Table 15 presents the complete comparison.[14]

Table 15. Blend performance on the same frozen weight protocol

Period	Allocation	Ann. return	Ann. vol	Sharpe	Max DD	5% 1m VaR
Training	Blend without output caps	3.79%	7.05%	0.395	-12.47%	3.35%
Training	Training-fitted capped ERC	3.08%	6.57%	0.315	-13.10%	3.12%
36-month test	Blend without output caps	13.91%	6.08%	1.537	-2.65%	2.89%
36-month test	Training-fitted capped ERC	12.74%	6.35%	1.286	-2.72%	3.01%

Source: BlackHawk Capital, own calculations and compilation. [14]

Table 16. ETF weights from the less constrained experiment

ETF	Training fitted blend	Current SAA
VOO	11.01%	9.91%
VTV	7.86%	10.54%
IEFA	7.81%	8.47%
EEM	7.59%	8.30%
IEF	15.01%	24.04%
IGSB	27.63%	12.00%
GLD	12.67%	11.75%
DBC	10.42%	15.00%

Source: BlackHawk Capital, own calculations and compilation. [14]

The improvement comes with 27.63% of capital in IGSB, more than twice our 12% cap. We retain the capped allocation because we prefer to distribute capital across the strategic exposures, while recognizing the stronger result of this tested alternative. The performance comparison uses capped ERC fitted in September 2023; the ETF-weight table compares the training-fitted blend with today's SAA. The references differ because they answer different questions.[14]

The holdout was selected during the research process, and the Sharpe difference has not been established as statistically significant. The experiment gives us evidence about a concentration trade-off, rather than proof of the best possible combination over all eligible portfolios.[14]

7.5 Covariance and window sensitivity

We challenge the weighting rule by changing its risk estimates. The 36-month, 120-month and full-history fits show how much the allocations depend on the historical window. The official formation estimate continues to use the final 60 months, as required by the guidelines. The volatility row makes a limitation explicit: the 60-month and 120-month fits land inside the 7-9% band at 8.6% and 7.2%, while the 36-month and full-history fits give 6.4% and 6.7%, below the 7% floor. Compliance with the risk budget is therefore a property of the prescribed estimation window as well as of the weights, and a team using a different window would have to re-optimize rather than simply re-price the same allocation.[14]

Table 17. Capped ERC under alternative estimation windows

ETF or vol	36m	60m	120m	167m
VOO	11.5%	9.9%	8.2%	8.6%
VTV	10.7%	10.5%	8.7%	9.1%
IEFA	9.3%	8.5%	7.3%	7.5%
EEM	8.2%	8.3%	7.0%	6.8%
IEF	23.9%	24.0%	33.0%	34.3%
IGSB	12.0%	12.0%	12.0%	12.0%
GLD	9.4%	11.7%	11.4%	10.4%
DBC	15.0%	15.0%	12.4%	11.3%
VOL %	6.4%	8.6%	7.2%	6.7%

Source: BlackHawk Capital, own calculations and compilation. [14]

Table 18. Sample and shrunk covariance weights

ETF	Sample covariance	Ledoit Wolf covariance
VOO	9.91%	10.18%
VTV	10.54%	10.79%
IEFA	8.47%	8.78%
EEM	8.30%	8.59%
IEF	24.04%	22.67%
IGSB	12.00%	12.00%
GLD	11.75%	12.00%
DBC	15.00%	15.00%

Source: BlackHawk Capital, own calculations and compilation. [14]

Ledoit-Wolf shrinkage moves the sample covariance towards a simpler target, with estimated intensity of 0.103. The largest instrument-weight change is 1.37 percentage points. Under the same sample covariance, the shrunk-weight allocation has estimated volatility of 8.68%, versus 8.58% for our chosen allocation. Evaluating the shrunk weights with their own shrunk covariance gives 8.33%. Keeping these estimates separate makes the sensitivity test interpretable.[14]

8 Costs liquidity and implementation

8.1 Fund expenses and transaction costs

The portfolio must be affordable to hold as well as coherent on paper. The eight-fund screen combines the reported AUM, fees and spreads with our weighted fee contributions. At current weights, the annual weighted net expense ratio is 28.56 basis points, or 0.2856%.[14]

Table 19. Eight fund implementation screen

ETF	AUM USD bn	Net fee per year	Average spread bps	Fee contribution bps
VOO	1000	0.03%	0.4	0.30
VTV	188	0.03%	1.0	0.32
IEFA	188	0.07%	1.0	0.59
EEM	30.7	0.72%	1.5	5.97
IEF	41.3	0.15%	1.1	3.61
IGSB	23.4	0.04%	2.0	0.48
GLD	140	0.40%	0.6	4.70
DBC	1.95	0.84%	3.1	12.60

Source: FactSet. [11]

DBC contributes 12.60 basis points and EEM 5.97 basis points to that total. These are the clearest places to review the implementation cost of our chosen exposures. The figures describe today's fund fees; historical fund expenses are already reflected in ETF returns and are not deducted again. Reported average spreads do not specify their averaging horizon.[14]

For the walk-forward trading-cost sensitivity, we charge 10 basis points for every dollar bought or sold. Turnover compares the new targets with holdings drifted by the preceding returns. Net return includes that assumed cost multiplicatively. Initial establishment costs are excluded. This gives a transparent sensitivity check, rather than a calibrated execution estimate.[14]

8.2 Implementation capacity

Capacity deserves particular attention in a USD 100 million portfolio. We estimate average daily traded dollars over the last 60 sessions through 9 October by multiplying each closing price by its reported share volume. This approximation gives a common basis for comparing our intended position sizes.[14]

Table 20. Secondary market capacity over 60 sessions

ETF	Mean USD m	Position USD m	Position / mean	Overview daily USD m
VOO	5,063	9.91	0.2%	3,000
VTV	598	10.54	1.8%	334
IEFA	869	8.47	1.0%	876
EEM	1,302	8.30	0.6%	976

Source: BlackHawk Capital calculations on FactSet data. [11][10][14]

Table 20. Secondary market capacity over 60 sessions (continued)

ETF	Mean USD m	Position USD m	Position / mean	Overview daily USD m
IEF	781	24.04	3.1%	593
IGSB	186	12.00	6.4%	231
GLD	3,733	11.75	0.3%	2,000
DBC	30	15.00	50.7%	21

Source: BlackHawk Capital calculations on FactSet data. [11][10][14]

DBC's USD 15 million position is 50.7% of the 60-session mean and 71.4% of the separate rounded USD 21 million single-day Overview value. The latter is not average volume. DBC is therefore our clearest execution-capacity concern. For AMC, all eight positions are entered on 17 or 18 October using the closing prices of 16 October, and remain unchanged until Round 2. The volume comparison flags potential execution costs in an institutional USD 100 million mandate; we do not assume those costs are negligible or introduce a staged acquisition into the challenge portfolio.[14][13]

The team interprets CVol as consolidated daily share volume. Closing price times shares approximates turnover, while actual traded value depends on transaction prices. The exports may differ in market coverage and rounding. ETF creation/redemption capacity could provide additional liquidity, but has not been modeled.[14]

Table 21. Walk forward sensitivity to assumed trading costs

Method	Gross Sharpe	Net Sharpe	Annual cost bps
EW	0.661	0.658	2.77
RP	0.587	0.583	2.72
ERC	0.718	0.712	4.35
ERC cap	0.583	0.579	3.17
MVO	0.691	0.680	10.72

Source: BlackHawk Capital, own calculations and compilation. [14]

The bond holdings also have distinct implementation characteristics: IEF and IGSB report durations of 6.8 and 2.7 years in their August snapshots. GLD holds physical gold; DBC uses commodity futures. We follow that explicit economic description for DBC where a generic derivatives flag in the export is inconsistent with it.[11][14]

8.3 Final portfolio quantities

Table 22 translates the weights into the executed share quantities. Rounding down to whole shares invests USD 99,999,453.61 and leaves USD 546.39 in cash. The effective annual volatility after rounding is 8.5753%.[14]

Table 22. Portfolio description: complete position record

Security name	Ticker	ISIN	Ccy	Shares	Weight	Price USD	Price date
Vanguard S&P 500 ETF	VOO	US9229083632	USD	14,123	9.91%	701.61	16 Oct 2026
Vanguard Value ETF	VTV	US9229087443	USD	47,173	10.54%	223.41	16 Oct 2026
iShares Core MSCI EAFE ETF	IEFA	US46432F8427	USD	87,735	8.47%	96.53	16 Oct 2026

Source: BlackHawk Capital, own calculations and compilation. [14]

Table 22. Portfolio description: complete position record (continued)

Security name	Ticker	ISIN	Ccy	Shares	Weight	Price USD	Price date
iShares MSCI Emerging Markets ETF	EEM	US4642872349	USD	132,047	8.30%	62.85	16 Oct 2026
iShares 7-10 Year Treasury Bond ETF	IEF	US4642874402	USD	265,843	24.04%	90.42	16 Oct 2026
iShares 1-5 Year Investment Grade Corporate Bond ETF	IGSB	US4642886463	USD	233,439	12.00%	51.40	16 Oct 2026
SPDR Gold Shares	GLD	US78463V1070	USD	30,755	11.75%	382.01	16 Oct 2026
Invesco DB Commodity Index Tracking Fund	DBC	US46138B1035	USD	444,531	15.00%	33.74	16 Oct 2026

Source: BlackHawk Capital, own calculations and compilation. [14]

Table 22 is the portfolio description required by the guidelines, with the security name, both identifiers, the currency, the quantity, the weight and the price in a single record. All eight trades are purchases and no position is sold, reduced or shorted. The prices are the closing prices of Friday 16 October 2026, and the share quantities are the ones entered in the Bloomberg portfolio on 17 October 2026.[14]

The quantities and their price date come from the Friday 16 October close. Invested value plus residual cash reconciles to USD 100 million, and the effective weights and risk were verified again at the executed quantities: every position is within 0.01 percentage points of its target weight. Appendix A supplies each fund's full identifiers and currency.[14]

The portfolio was entered on 17 October 2026 using PRTU, equity portfolio type and shares/par amount, as BLACKHAWK CAPITAL, identifier U33651819-1, denominated in USD. Corporate-action settings adjust cash for coupons and dividends and reinvest cash dividends. The Bloomberg positions match this report and are shared with the teaching assistant; they remain unchanged until Round 2.[13]

The acceptable maximum drawdown and the realistic discounted Sharpe required by the checkpoints are set at 20% and 0.29. The quantities above are stated on the 16 October closes and match the shared Bloomberg portfolio. The report, code and portfolio description are due on 18 October at 23:59 and must be uploaded to the fund website.[13]

Appendix A Instrument identifiers and sources

Table 23. Full instrument identifiers

ETF	ISIN	Currency
VOO	US9229083632	USD
VTV	US9229087443	USD
IEFA	US46432F8427	USD
EEM	US4642872349	USD
IEF	US4642874402	USD
IGSB	US4642886463	USD
GLD	US78463V1070	USD
DBC	US46138B1035	USD

Source: FactSet. [11]

Portfolio returns and quantities use the team's FactSet Price History exports: Gross Total Return, raw Price and CVol. Overview PDFs supply AUM, net expense ratios, average spreads and fund characteristics. Holdings, Sector and Country Portfolio Trend exports provide the composition inputs. Figures and comparative tables in this report are BlackHawk Capital's own calculations or compilations from those inputs. Prior-year Round 1 reports from other teams were reviewed as format references only; they contributed no data, estimates or calculations to this report.[10][11][12][14][15]

The daily market-data cutoff is 9 October 2026 and the performance cutoff is September 2026. AUM and trading statistics are dated 9 October. US holdings and sectors use August snapshots; IEFA and EEM use September. Country comparisons use a common September snapshot. Expense ratios are not separately dated, and the spread averaging horizon and precise country-attribution rule are absent from the exports.[11][12]

Risk-free input: the team's daily FRED DGS1MO download, 1 October 2012-30 September 2026, converted to monthly yields as described in Section 3. The professor's file is retained for comparison. Course series supply TIP, LQD, SPY and EFA for alternative-universe tests; the other tested price histories are team FactSet exports.[9][16][14]

The source audit passed 188 numeric checks against the supplied exports and the notebook reproduced all 31 principal report measures. Provider terminology that is not documented remains identified as such. The full diagnostics and 120 notebook outputs are retained in the editable HTML companion and audit files.[14]

Appendix B Code and data map

Table 24. Core analytical files

File	Purpose
SAA_Round1_analysis.ipynb	Main English analysis, allocation, backtests, tests and exhibits; run sequentially.
costruisci_csv.py	Prepare aligned price CSVs from original Excel exports.
etf_adj_prices.csv / etf_close_prices.csv	Daily total-return and raw-close panels for the eight ETFs.
rf_dgs1mo_daily.csv / rf_team_monthly.csv	Team FRED source and derived active monthly yield proxy.
candidates_adj_prices.csv / substitutes_adj_prices.csv	Aligned candidate panels; some columns originate from course data, others from FactSet.
equity holdings sector and country CSVs	Parsed source snapshots supporting composition and overlap.
fund_overview_inputs.csv / provenance CSV	PDF-derived implementation fields and field-level source references.
research_variants_register.csv / report_policy_inputs.json	Documented research attempts and explicit policy/source choices.
portfolio_for_PRTU.csv	Executed portfolio description: shares, weights and the 16 October 2026 closing prices entered in Bloomberg PRTU.
exhibits folder	Saved figures and source tables for the report.

Source: BlackHawk Capital, own calculations and compilation. [14]

The attached notebook is the main analytical record. It loads the aligned return histories and the team's risk-free series, estimates the inputs, fits the allocation methods, runs the historical tests, and exports the composition and implementation exhibits. The report presents those results.[14]

The HTML companion retains all 120 notebook outputs under Complete outputs, including the extended tables and diagnostic plots omitted from the printable report. The research register, source checks and editable exhibit CSVs accompany the code.[14]

Appendix C Research configurations

We record 34 economic configurations and protocol attempts. The list includes the five main allocation methods, sensitivity fits, ETF additions and substitutions, early documented blend trials and the optimized blend experiments. Earlier trials are labeled where they were not recomputed in the final notebook. The register is a minimum documented count; it does not enumerate every numerical iteration.[14]

Table 25. Documented research history

Configuration	Category	Evidence
Equal Weight	Allocation method	recomputed in current notebook
Inverse-volatility Risk Parity	Allocation method	recomputed in current notebook
Uncapped ERC	Allocation method	recomputed in current notebook
Capped ERC	Allocation method	recomputed in current notebook
Constrained Maximum Sharpe	Allocation method	recomputed in current notebook
Capped Minimum Variance diagnostic	Allocation method	recomputed in current notebook
Capped ERC with Ledoit-Wolf covariance	Covariance estimator	recomputed in current notebook
Capped ERC: 36 months	Estimation-window sensitivity	recomputed in current notebook
Capped ERC: 120 months	Estimation-window sensitivity	recomputed in current notebook
Capped ERC: full common history	Estimation-window sensitivity	recomputed in current notebook
Add TIP	Universe addition	recomputed in current notebook
Add LQD	Universe addition	recomputed in current notebook
Add EMB	Universe addition	recomputed in current notebook
Add USMV	Universe addition	recomputed in current notebook
Add VNQ	Universe addition	recomputed in current notebook
VOO -> SPY	ETF substitution	recomputed in current notebook
IEFA -> EFA	ETF substitution	recomputed in current notebook
EEM -> VWO	ETF substitution	recomputed in current notebook
VTV -> USMV	ETF substitution	recomputed in current notebook
IEF -> TLT	ETF substitution	recomputed in current notebook
IGSB -> LQD	ETF substitution	recomputed in current notebook
IGSB -> TIP	ETF substitution	recomputed in current notebook
IGSB -> EMB	ETF substitution	recomputed in current notebook
GLD -> IAU	ETF substitution	recomputed in current notebook
Add TIP and LQD together	Universe addition	Earlier documented trial; not recomputed in current notebook
Add USMV and VNQ together	Universe addition	Earlier documented trial; not recomputed in current notebook

Source: BlackHawk Capital, own calculations and compilation. [14]

Table 25. Documented research history (continued)

Configuration	Category	Evidence
50% capped ERC + 50% Minimum Variance	Fixed method blend	Earlier documented trial; not recomputed in current notebook
one-third capped ERC / Risk Parity / Minimum Variance	Fixed method blend	Earlier documented trial; not recomputed in current notebook
50% capped ERC + 50% Equal Weight	Fixed method blend	Earlier documented trial; not recomputed in current notebook
70% capped ERC + 30% Minimum Variance	Fixed method blend	Earlier documented trial; not recomputed in current notebook
Optimized six-method blend with final 36-month holdout	Optimized blend	recomputed in current notebook
Optimized blend with initial half-history split	Protocol attempt	infeasible under imposed constraints
Optimized six-method blend without project instrument caps	Optimized blend	recomputed in current notebook
Maximum Sharpe without volatility band (risk-return diagnostic)	Allocation diagnostic	recomputed in current notebook

Source: BlackHawk Capital, own calculations and compilation. [14]

References

External sources accessed on 12 October and rechecked in the audit completed on 13 October 2026. The report information cutoff remains 12 October. Local exports retain their field dates. Forecasts are attributed to their issuing institution; portfolio interpretations and calculations are BlackHawk Capital's own.

[1] Federal Reserve. *FOMC statement*. 16 September 2026. https://www.federalreserve.gov/newsevents/pressreleases/monetary20260916a.htm

[2] Federal Reserve. *Summary of Economic Projections*. 16 September 2026. https://www.federalreserve.gov/monetarypolicy/fomcprojtabl20260916.htm

[3] US Bureau of Labor Statistics. *Consumer Price Index August 2026*. 11 September 2026. https://www.bls.gov/news.release/archives/cpi_09112026.htm

[4] European Central Bank. *Monetary policy decisions and staff projections*. 10 September 2026. https://www.ecb.europa.eu/press/pr/date/2026/html/ecb.mp260910~314e508016.en.html

[5] International Monetary Fund. *World Economic Outlook Update*. 8 July 2026. https://www.imf.org/en/publications/weo/issues/2026/07/08/world-economic-outlook-update-july-2026

[6] International Monetary Fund. *July World Economic Outlook press briefing*. 8 July 2026. https://www.imf.org/en/news/articles/2026/07/08/tr070826-weo-press-briefing-transcript-july-8-2026

[7] US Energy Information Administration. *Short Term Energy Outlook October 2026*. 6 October 2026. https://www.eia.gov/outlooks/steo/archives/oct26.pdf

[8] World Gold Council. *Gold Demand Trends Q2 2026*. 30 July 2026. https://www.gold.org/goldhub/research/gold-demand-trends/gold-demand-trends-q2-2026

[9] Federal Reserve Bank of St Louis. *DGS1MO via FRED*. Team daily export through 30 September 2026. https://fred.stlouisfed.org/series/DGS1MO

[10] FactSet. *ETF Price History exports*. Through 9 October 2026. Team Gross Total Return, raw Price and CVol inputs.

[11] FactSet. *Eight ETF Overview exports*. 12 October 2026; field dates disclosed separately. Team PDFs in factset_screening.

[12] FactSet. *ETF composition exports*. August and September 2026. Team holdings, sector and country inputs; conventions in Appendix A.

[13] USI Financial Modeling. *AMC Round 1 and General guidelines*. 2026. Supplied course PDFs.

[14] BlackHawk Capital. *Portfolio calculations and research register*. Audited 13 October 2026. Team Python notebook, exhibit data and source audit in the accompanying technical files.

[15] Aura Capital WhiteRock Capital Apex Lugano and PIG Capital. *Round 1 reports*. 2025. Supplied reports used as presentation references.

[16] USI Financial Modeling course data. *Adjusted-price course dataset*. SPY, EFA, LQD and TIP. Retained copy: report/audit/source_inputs/course_adj_prices.csv.

[17] Lo, A. W.. *The Statistics of Sharpe Ratios*. Financial Analysts Journal 58(4), 36-52. 2002. DOI 10.2469/faj.v58.n4.2453. Equation 9 gives the IID standard error used in Section 4.5.

[18] Jobson, J. D. and Korkie, B. M.. *Performance Hypothesis Testing with the Sharpe and Treynor Measures*. Journal of Finance 36(4), 889-908. 1981. Paired test for the difference between two Sharpe ratios.

[19] Memmel, C.. *Performance Hypothesis Testing with the Sharpe Ratio*. Finance Letters 1, 21-23. 2003. Correction to the Jobson-Korkie statistic, applied in Section 4.5.

[20] Ledoit, O. and Wolf, M.. *Robust Performance Hypothesis Testing with the Sharpe Ratio*. Journal of Empirical Finance 15, 850-859. 2008. https://www.ledoit.net/jef_2008pdf.pdf

[21] Michaud, R. O.. *The Markowitz Optimization Enigma: Is 'Optimized' Optimal?*. Financial Analysts Journal 45(1), 31-42. 1989. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2387669

[22] DeMiguel, V., Garlappi, L. and Uppal, R.. *Optimal Versus Naive Diversification: How Inefficient Is the 1/N Portfolio Strategy?*. Review of Financial Studies 22(5), 1915-1953. 2009. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=1376199

[23] Franzoni, F., USI Financial Modeling. *Round 1 checkpoint assignments and backtest interpretation*. 2026 in-class slides. Class 3 Part I slide 26; Class 3 Part II slide 23; Class 4 slides 15-16. Supplied course PDFs.

[24] Franzoni, F., USI Financial Modeling. *Class 2 performance measures*. 2026 lecture notes. Supplied Class 2 reading; Sortino definition on PDF page 17.

[25] US Bureau of Labor Statistics. *Consumer Price Index September 2026*. 14 October 2026. https://www.bls.gov/news.release/archives/cpi_10142026.htm

[26] US Bureau of Labor Statistics. *Producer Price Index September 2026*. 15 October 2026. https://www.bls.gov/news.release/archives/ppi_10152026.htm